

Measuring Bank Exposure to Oil Shocks: Variables, Data Sources, and Method

1 Variables in the current data worth using

The data contain no direct energy-loan variable, because the Call Report does not break C&I loans out by industry. Exposure must therefore be proxied by *where* a bank operates and *what* it lends.

Exposure (who is exposed)

- **Geography:** State Code, State and County Number, Zip, City, CBSA Name, FDIC Region, Federal Reserve District. This is the strongest identifier because oil exposure is local; it links to county-level oil dependence.
- **Loan mix:** Commercial & Industrial Loans, Construction and Development, Non-Farm Non-Residential (CRE), Multi-Family Residential, 1–4 Family Residential, Loans to Individuals, Total Real Estate Loans. Scale each by total assets or total loans.
- **Funding mix:** Non-Interest-Bearing Deposits, Core (Retail) Deposits, Estimated Uninsured Deposits, Total Deposits, Net Loans to Deposits.

Outcomes (how they respond)

- Net Charge-Offs to Loans; Non-Current Loans to Loans; Non-Current Assets plus OREO to Assets
- Provision to Assets; Loss Allowance to Loans
- Return on Assets; Net Interest Margin; Net Loans growth; Other Real Estate Owned

Controls

log(Total Assets), Equity Capital to Assets, Tier 1 Risk-Based Capital Ratio, securities share of assets, Efficiency Ratio, Institution Class, Multi-Bank Holding Company and holding-company ID (for clustering), Trust Powers.

Skip

Dates, FDIC office and regulator codes, Form Number, and duplicate ratio variants. Keep either the dollar level or the ratio, not both.

2 Data sources that join easily

Source	Adds	Join key
FDIC Summary of Deposits (SOD)	Branch-level deposits by county, annual; essential for weighting exposure	Cert, county FIPS
BLS QCEW	County employment and wages in oil and gas (NAICS 211, 213111/2, 2111)	County FIPS
BEA Regional EIA	County or state GDP from mining and oil and gas	County FIPS
FRED	State and county oil production, reserves	State or county
	WTI (DCOILWTICO), Brent, state unemployment, macro controls	Date
Baker Hughes	Rig counts by state	State, date
FFIEC bulk Call Reports	Full schedules, e.g. RC-C loans by collateral and size	RSSD ID
CRA and HMDA	Small-business and mortgage lending by county	Tract or county
CRSP	Bank stock returns, a market-based check for public banks	NY Fed RSSD–PERMCO link

SOD and QCEW are the most valuable additions: together they give a bank-level, geographically weighted exposure.

3 Suggested method

Step A: a predetermined exposure index (shift-share style)

Weight each county’s oil intensity by the bank’s deposit footprint:

$$\text{Exposure}_i = \sum_c \text{DepositShare}_{i,c} \times \text{OilIntensity}_c.$$

OilIntensity_c is oil and gas employment (or mining GDP) as a share of total county employment. Fix both the weights and the intensity at a **pre-period** (e.g., 2010–2013) so banks’ own choices during the shock do not contaminate the measure. Optionally interact the index with loan mix (e.g., the C&I plus CRE plus construction share) to capture banks that are both located in oil counties and lend heavily to cyclical sectors.

Step B: a behavioral sensitivity (bank-specific beta)

For each bank estimate

$$y_{it} = \alpha_i + \beta_i \Delta \log(\text{WTI})_{t-k} + \gamma' X_{it} + \varepsilon_{it},$$

where y is ROA, the NCO ratio, or loan growth, and k ranges over 0–4 quarters. β_i is the bank’s realized sensitivity. Rolling windows work but are noisy with few quarters, so shrink the betas (empirical Bayes) or estimate them by group.

Step C: panel test tying the two together

$$y_{it} = \alpha_i + \lambda_t + \delta (\text{Exposure}_i \times \Delta \text{Oil}_t) + \gamma' X_{it} + \varepsilon_{it}.$$

Time fixed effects λ_t absorb the oil price itself, so δ is identified from cross-sectional differences in exposure. Run it on the 2014–16 collapse and on 2020, and separate up moves from down moves

since effects are likely asymmetric. Cluster standard errors by bank (or state, if exposure varies mostly at that level).

Step D: validation

- Check that the Step A exposure predicts the Step B betas.
- Compare with equity-return betas for public banks.
- Run a placebo test on non-oil counties or periods.

4 Connection to the MLE model

The exposure work above supplies the inputs $x_{it} = (E'_i, w'_{i,t-h})'$ that the distributional MLE in `mle_proposal.tex` interacts with the oil supply shock s_t . The MLE also replaces two pieces of the regression design in Section 3.

Where each piece enters

This document	Role in the MLE
Step A exposure index	The producer-side element of E_i , entering every index η_{it}^m ($\mu, \sigma, \epsilon, \delta$) through x_{it} and $s_t x_{it}$
Consumer-side index (same construction, oil-intensive NAICS such as transportation and manufacturing in place of oil and gas)	The consumer-side element of E_i ; the sign of the response differs by side, so the two are kept separate
Loan-mix shares (C&I, CRE, construction)	Optional additional elements of E_i , or lagged elements of $w_{i,t-h}$ if measured at $t - h$
Equity to assets, securities and cash share, core-deposit share, log assets	The lagged characteristics $w_{i,t-h}$, standardized on the training sample
Step C panel regression	The mean equation of the Gaussian baseline; δ in Step C corresponds to h_μ
Step B bank-specific beta	Replaced by the model's implied response $\partial Q_p / \partial s$ evaluated at bank i 's x_{it} , so sensitivity is reported by quantile, not as one number

Changes needed to make the two consistent

1. **Shock.** Use the identified supply shock s_t (Kilian, Baumeister–Hamilton, or Känzig), not $\Delta \log(\text{WTI})$. Keep $\Delta \log(\text{WTI})$ as the demand-driven contrast in the identification checks.
2. **Outcomes.** Use the MLE outcomes at $h = 4$: $100 [\ln A_{it} - \ln A_{i,t-4}]$ and four-quarter changes in equity ratio, ROA, and NIM. NCO, non-current, and provision ratios are non-negative and heavily right-skewed; they need an arcsinh transform (or a separate model) before the sinh-arcsinh family is a sensible fit.

3. **Exposure window.** The MLE requires E_i to be fixed at or before the start of the sample. Summary of Deposits begins in 1994, so weights fixed at 1994 imply starting the estimation sample in about 1995 rather than 1992. QCEW and BEA county data are available for the pre-period.
4. **Scaling.** Deposit-weighted oil intensity is zero or near zero for most banks and very large for a few. Standardize an arcsinh or log transform of the index (the simulation in `src/simulate_mle.py` uses a standard-normal E_i), and report results by exposure quantile so the interaction is not driven by a handful of Texas, Oklahoma, and Louisiana banks.
5. **Time effects.** Quarter effects absorb s_t , so as in the MLE the aggregate effect b_m comes from the specification without them and the exposure interactions h_m from the one with them.

What the exposure measure lets the MLE test

- $h_\mu \neq 0$: high-exposure banks have a different mean response, the Step C result.
- $h_\sigma \neq 0$: the shock widens dispersion more for exposed banks.
- $h_\epsilon \neq 0$ or $h_\delta \neq 0$: the shock moves the *tails* of exposed banks differently. This is the new information over a mean regression, and it is what the quantile-response curves $\partial Q_p / \partial s$, drawn separately for high- and low-exposure banks, display.
- Producer versus consumer coefficients of opposite sign, consistent with a supply shock that raises prices, help produce mechanism evidence.

Suggested order of work

1. Build the producer and consumer indices from SOD and QCEW, fixed at 1994.
2. Merge them and the lagged controls into the bank-quarter panel with outcomes at $h = 4$.
3. Replace the standard-normal E_i in `simulate_mle.py` with the empirical index distribution to check that the SHASH estimator recovers known parameters when exposure is this skewed.
4. Estimate the Gaussian baseline, then the SHASH model, with quarter-clustered errors, followed by the quantile-response figures and placebo test on pre-shock quarters.

5 Caveats

- Exposure from deposit geography measures where deposits are gathered, not necessarily where loans are made; CRA/HMDA lending by county can cross-check this.
- Large banks with national footprints show diluted exposure; consider size buckets or excluding the largest banks.
- Oil prices coincide with broader macro shocks (2008, 2020); include state-level macro controls or year-by-region fixed effects.